

(1)

① $a \geq 0$, $[a] = a+1$

② $a < 0$, $[a] = -2a-1$



$[1] = 1+1=2$

$a=1, a \geq 0$ $[1] = 1+1=2$

$[-1] = -2 \times (-1) - 1 = 1$

$= 2-1=1$

$[a] = 0$

$a+1=0, a=-1$

(X)

$-2a-1=0, -2a=1$

$a = -\frac{1}{2}$ ✓

(2)

$\begin{cases} [m-n] = 0 \\ [m+n] = 2 \end{cases}$

① $m-n \geq 0$

$[m-n] = (m-n)+1 = m-n+1 = 0$

$[m+n] = m+n+1 = m+n+1 = 2$

$(m-n+1) + (m+n+1) = 0+2$

$2m+2=2$

$m+1=1$

$m=0$

$|m-n+1| = 0$

$0-n+1=0$

$n=1$

$\begin{cases} m=0 \\ n=1 \end{cases}$

② $m-n < 0$

$[m-n] = -2(m-n)-1=0$

$2(m-n) = -1$

$2m-2n+1=0$

$[m+n] = m+n+1=2$

$2m+m+2=4$

$2m-2n+1 + 2m+m+2 = 0+4$

$4m-2n+3=4$

$4m=1$

$m = \frac{1}{4}$

$m+n+1=2$

$n = \frac{3}{4}$

$\begin{cases} m = \frac{1}{4} \\ n = \frac{3}{4} \end{cases}$

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④

$$5x + 6y = 50$$

$$5x = 50$$

$$x = 10$$

$$y = \frac{50}{6} \leq 8$$

① $x = 10, y = 0$

$$x = 9, y = \frac{50 - 5x}{6} = \frac{50 - 45}{6} = \frac{5}{6}$$

$$x = 8, y = \frac{50 - 40}{6} = \frac{10}{6} \text{ (X)}$$

$$x = 7, y = \frac{50 - 35}{6} = \frac{15}{6} \text{ (X)}$$

$$x = 6, y = \frac{50 - 30}{6} = \frac{20}{6} \text{ (X)}$$

$$x = 5, y = \frac{25}{6}$$

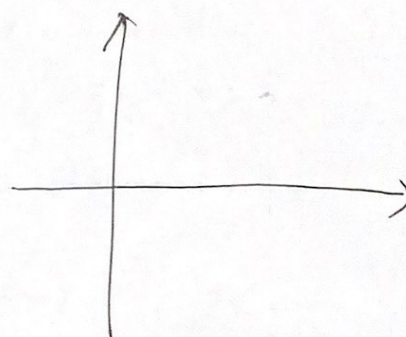
② $x = 4, y = \frac{30}{6} = 5$

$$x = 3, y = \frac{35}{6} \text{ (X)}$$

$$x = 2, y = \frac{40}{6} \text{ (X)}$$

$$x = 1, y = \frac{45}{6} \text{ (X)}$$

$$x = 0, y = \frac{50}{6} \text{ (X)}$$



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$$\begin{cases} 3x+5y=2a+1 & (1) \\ 2x+5y=4-a & (2) \end{cases}$$

(1) - (2)

$$3x+5y-(2x+5y)=2a+1-(4-a)$$

$$3x+5y-2x-5y=2a+1-4+a$$

$$x=3a-3 \quad (3)$$

(3) 代入 (2)

$$2x+5y=4-a$$

$$2(3a-3)+5y=4-a$$

$$6a-6+5y=4-a$$

$$6a-6-4+a=-5y$$

$$7a-10=-5y \quad 5y=10-7a$$

$$y=\frac{10-7a}{5}$$

$$2y=\frac{20-14a}{5}$$

$$x+2y=3a-3+\frac{20-14a}{5}$$

$$=\frac{15a-15}{5}+\frac{20-14a}{5}$$

$$=\frac{15a-15+20-14a}{5}$$

$$=\frac{5-a}{5}$$

$$2x-ky$$

$$=2(3a-3)-k(2-\frac{7}{5}a)$$

$$=6a-6-2k+\frac{7a}{5}k$$

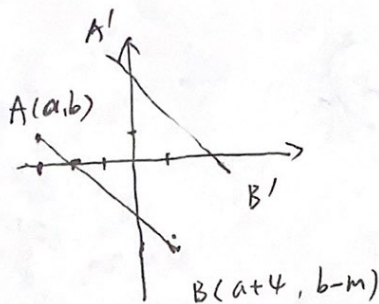
$$=(6+\frac{7}{5}k)a-6-2k$$

$$6+\frac{7}{5}k=0$$

$$30+7k=0$$

$$k=-\frac{30}{7}$$

☆☆ (17)



$A(a, b)$
 向右平移4个单位 $(a+4, b)$
 向下平移 $m(m>0)$ $(a+4, b-m)$
 B点

AB向右平移3个单位

$A_1(a+3, b)$
 $B_1(a+7, b-m)$

AB向上平移 $n(n>0)$

$A'(a+3, b+n)$
 $B'(a+7, b-m+n)$

分析坐标过程

$$(1) |a+3| + \sqrt{b-1} = 0, n=2$$

$$|a+3|=0, \sqrt{b-1}=0$$

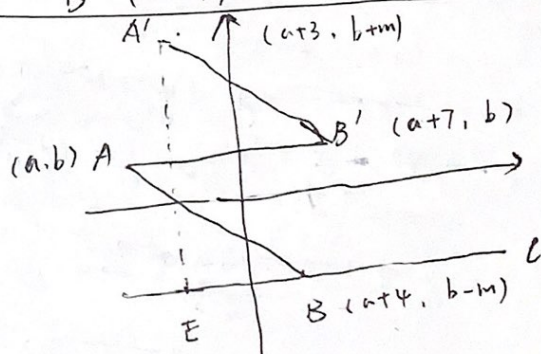
$$a=-3, b=1$$

$$A'(a+3, b+n)$$

$$a+3 = -3+3=0$$

$$b+n = 1+2=3$$

$$A'(0, 3)$$



$$\text{Min}(A'E) = 8$$

$$A'E \perp BE$$

$$AB' \parallel BE \parallel x\text{轴}$$

(2) ① 若 $AB' \parallel x\text{轴}$

~~A~~

$$y_A = y_{B'}$$

$$b = b-m+n$$

$$m=n$$

$$A(a, b)$$

$$B(a+4, b-m)$$

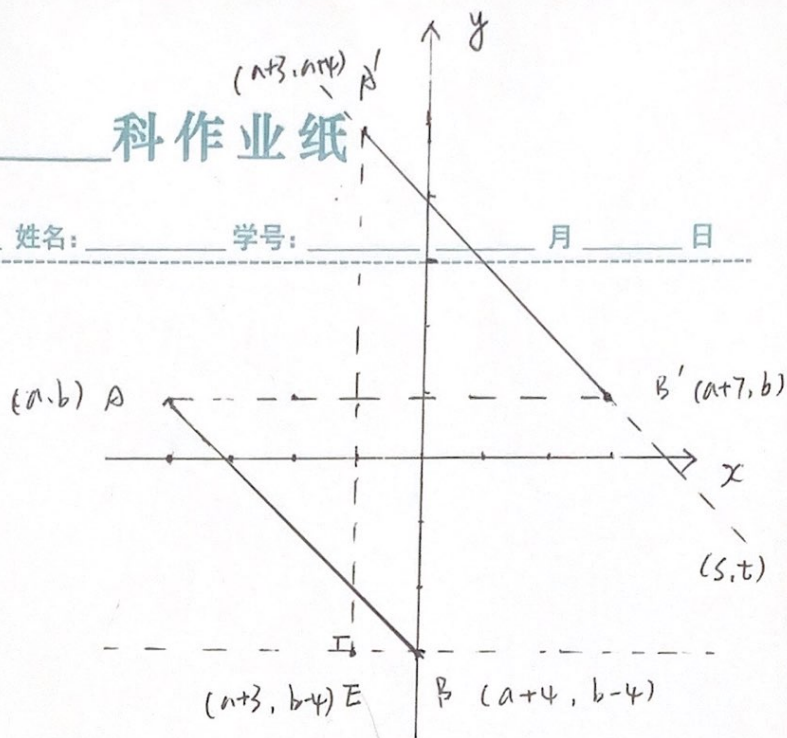
$$A'(a+3, b+n) = (a+3, b+m)$$

$$B'(a+7, b-m+n) = (a+7, b)$$

$$b+m - (b-m) = A'E = 8$$

$$b+m - b + m = 2m = 8$$

$$m=4, n=4$$

$A(a, b)$
$$B(a+4, b-4)$$
$$A' (a+3, b+4)$$
$$B'(a+7, b)$$


$$A'E = 8$$

$A', B', (s, t) \quad \boxed{px + qy = k} \quad (\cancel{pq} \neq 0)$

(s, t) 设为点 F, A', B', F 三点一线

判断 $s-a+t-b$ ($s \neq m$) 的值是否随着 s, t 的变化而变化

$$A' (a+3, b+4) \quad B' (a+7, b) \quad F (s, t)$$

$$\begin{cases} p(a+3) + q(b+4) = k & (1) \\ p(a+7) + q \cdot b = k & (2) \\ ps + qt = k & (3) \end{cases}$$

$$p(a+3) + q(b+4) - p(a+7) - q(b) = k - k = 0$$

$$p(a+3 - a-7) + q(b+4 - b) = 0$$

$$-4p + 4q = 0$$

$$4p = 4q$$

$$p = q \neq 0 \text{ (4)}$$

$$ps + qt = k = p(s+t) = k \quad \text{if } \Delta \textcircled{3}$$

$$p(a+3) + q(b+4) = p(a+b+7) = k \quad (\text{代入})$$

$$\hookrightarrow s+t = \frac{k}{p}$$

$$s+t = a+b+7$$

$$|a+b+7| = \frac{k}{p}$$

$$s+t-a-b=7$$